and the responsible person should:
.1 where it is necessary for shore personnel to enter any such spaces, conduct a pre-operational risk assessment prior to commencement of cargo operations or arrival of shore personnel. When hazards are identified a joint risk assessment with terminal representatives responsible for operations on board the ship, or with other appropriate shore personnel, should be conducted; and
.2 identify and agree to the precautions required during entry and enclosed space entry permit arrangements to be used.
6.3 Entry doors or access hatches leading to enclosed spaces should at all times be secured against entry unless the spaces have been risk assessed, atmospherically tested as required and declared safe for entry. The ship may use a system of seals similar to t hose
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required and declared safe for entry. The ship may use a system of seals similar to t hose
recommended in the ISPS Code. Suitable portable signage, which is easily understandable by the vessels crew and also by shore personnel engaged on the ship at this time and in the port, indicating the hazards should be posted on entry doors or access hatches leading to an enclosed space. An example of such signage is contained within appendix 3. These signs should be updated when the space becomes safe for entry or when a safe space becomes unsafe.
6.4 A door or hatch cover which is opened to provide natural ventilation of an enclosed space may, wrongly, be taken to be an indication of a safe atmosphere and therefore it is recommended to station an attendant at the entrance or use a mechanical or physical barrier, such as a locked bar or chain positioned across the opening with an attached warning sign, to
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such as a locked bar or chain positioned across the opening with an attached warning sign, to
prevent accidental entry. It is recommended that operations such as these are included in the watch handover activity.

6.5 The master or the responsible person should determine that it is safe to enter an enclosed space by ensuring that:
.1 potential hazards have been identified in the risk assessment and as far as possible isolated or made safe;
.2 the space has been thoroughly ventilated by natural or mechanical means to remove any toxic or flammable gases identified in the hazard identification process gases;
.3 the atmosphere of the space has been tested using certified and calibrated instruments to ascertain that the space contains 20.9% oxygen. If the testing
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instruments to ascertain that the space contains 20.9% oxygen. If the testing indicates that the level of oxygen is less than 20.9%, or indicates the presence of even low levels of flammable or toxic gases, then the space should be treated as one where the atmosphere is known or suspected to be unsafe, as described in section 9;
.4 the space has been declared safe for entry and properly illuminated;
.5 a suitable system of communication between all parties for use during entry has been agreed and tested, and the evacuation signals have been agreed upon;
.6 personnel entering the space are wearing personal gas detection equipment that has been properly calibrated and is capable of monitoring the levels of oxygen, carbon monoxide and any other gases identified in the risk assessment;
.7 an attendant has been nominated and properly instructed;
.8 rescue and resuscitation equipment has been positioned , tested and ready
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.7 an attendant has been nominated and properly instructed;
.8 rescue and resuscitation equipment has been positioned , tested and ready for immediate use at the entrance to the space and a rescue plan detailing the rescue arrangements has been agreed. Where ship or shore personnel are working in more than one cargo space at the same time, the rescue equipment should be positioned at a designated central location. In the event of an emergency in any one space, all personnel working in other spaces should be instructed to stop work immediately and exit the space;
.9 personnel are provided with the appropriate personal protective equipment for the entry and subsequent tasks; and
.10 the required permits have been issued, authorizing entry.
6.6 The precautions in sub -paragraphs .6 to .8 of paragraph 6. 5 above may not be applicable to every situation described in this section. The responsible person authorizing entry
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applicable to every situation described in this section. The responsible person authorizing entry should determine whether personal gas detectors, an attendant and the positioning of rescue equipment at the entrance to the space are necessary.
6.7 Only trained and authorized personnel should be assigned the duties of entering, functioning as attendants or acting as members of rescue teams. Ships' crews with rescue and first aid duties should be drilled periodically as required by SOLAS regulation I II/19.3.6 in rescue and first aid procedures. Training should include as a minimum:

.1 identification of the hazards likely to be faced during entry into enclosed spaces and whilst within the space, in particular the rapidity with which
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spaces and whilst within the space, in particular the rapidity with which oxygen may be depleted in a space by corrosion or biological means especially in higher ambient temperatu res. The pace at which oxygen gets depleted increases exponentially when ventilation is stopped (fan or other equipment in case of forced ventilation, and closing a vent or damper in case of natural ventilation) or other characteristics or properties of the contents or structure of the space. While higher ambient temperatures can increase the rate of oxygen depletion, it should be assumed that the rate of depletion will always be very rapid, regardless of the ambient temperature;
.2 an explanation of how the pace at which oxygen is depleted will increase exponentially when ventilation is stopped (fan or other equipment in case of forced ventilation, and closing a vent or damper in case of natural ventilation), or when hatch covers are closed;

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forced ventilation, and closing a vent or damper in case of natural ventilation), or when hatch covers are closed;
.3 identification and use of the various sources of information on the hazards associated with individual solid bulk and liquid bulk cargoes, and the precautions to be adopted when entering spaces containing such cargoes, or their residues;
.4 awareness of the fact that when a person or persons in an enclosed space shows signs of adverse health effects, that they should always assume that these effects are due to an oxygen -depleted or toxic atmosphere in the space, and that they should not enter it themselves;
.5 recognition of the signs of adverse health effects caused by exposure to hazards during entry;
.6 knowledge and experience in the use of personal protective equipment required for entry;
.7 rescue, first aid, Cardio Pulmonary Resuscitation (CPR) techniques and evacuation procedures;

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required for entry;
.7 rescue, first aid, Cardio Pulmonary Resuscitation (CPR) techniques and evacuation procedures;
.8 knowledge and experience in the use of communication equipment and procedures;
.9 knowledge and experience in the use of instruments for measuring the atmosphere;
.10 knowledge and experience in the use of rescue equipment and procedures;
.11 knowledge of the IMO/WHO/ILO Medical First Aid Guide for Use in Accidents involving Dangerous Goods (MFAG), where appropriate; and
.12 knowledge of use of emergency and the first aid equipment for chemical

tankers (IBC Code section 14.3) and for gas carriers (IGC Code
section 14.3), where appropriate.
6.8 All equipment used in connection with entry should be in good working condition and
inspected and tested prior to use.

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REVISED RECOMMENDATIONS FOR ENTERING
ENCLOSED SPACES ABOARD SHIPS
MSC 110/21/Add.3
Annex 28, page 11
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6.9 As far as practicable, enclosed space entry should be carried out during hours of
daylight or normal working hours of the ship to ensure ready availability of reserve personnel
in the event of an emergency.
7 TESTING THE ATMOSPHERE
7.1 The gas detection equipment should be appropriate for the cargo that the ship has
carried and is carrying considering information provided in the shipper's declaration, the Safety
Data Sheets (SDS), the IMDG Code, the IMSBC Code, International Bulk Chemical Code (IBC
Code) and the IGC Code. The gas detection equipment, including devices for testing CO 2,
should be capable of operating correctly even in oxygen-depleted atmosphere.
7.2 Appropriate testing of the atmosphere of a space should be carried out with properly
calibrated equipment by persons trained in the use of the equipment. The manufacturers'

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calibrated equipment by persons trained in the use of the equipment. The manufacturers'
instructions should be strictly followed. Testing of the atmosphere of a space should be carried
out before any person enters the space and at regular intervals thereafter until all work is
completed. Where appropriate, the testing of the atmosphere of a space should be carried out
at as many different levels and areas as is necessar y to account for gas stratification, and
obtain a representative sample of the atmosphere in the space. In some cases, it may be
difficult to test the atmosphere throughout the enclosed space without entering the space